

EV Adoption in Washington State: Market Composition and Adoption Drivers

Alessandro Felici

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1 Introduction

The data set used is “Electric Vehicle Population Data” produced by the Research and Analysis Office in the Department of Licensing from the State of Washington. The dataset tracks all battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs) currently registered in Washington State. It contains vehicle manufacturer information, electricity usage, and location data. It also includes eligibility for Clean Alternative Fuel Vehicle credits, a temporary incentive for using clean energy vehicles. CAFV eligibility identifies vehicles qualifying for tax exemptions based on clean fuel and environmental standards, specifically for electric-only vehicles. Limitations were that the data set is a snapshot of registration, not a change in records over time. We cannot investigate when purchases occurred. It is limited to one state, and not generalizable to the country or other countries. Additionally, there is a lack of data for BEV electric range in recent model years.

This project aimed to answer the following question: How is the electric vehicle market changing in Washington State? To quantify this question, I decided to break it down into the following four components: Question 1: Which electric vehicle manufacturers and models are most common in the state of Washington? We expect brands such as Ford, Toyota, and Tesla to appear often, with a particular emphasis on Tesla. Question 2: How has the combination of BEVs vs PHEVs changed for each year of production? We expect the share of BEVs to grow each year, and the PHEV to react to this growth in EV usage. Question 3: Does electric range differ significantly between BEVs and PHEVs, and has it improved across model years? From this, we expect PHEV range to improve to some degree over model years. Then, additional research from these base sub questions was done through Question 4: What can be inferred about the EV data by zip code area? Can we predict usage by demographic information? From this, I was able to find numeric evidence which can derive an analysis about electric vehicle data in the State of Washington. In these parts, we expect to find some correlation between demographic traits, and discover geography to have some ability to predict EV type.

2 Methods

In question 1 we extracted the number of vehicles for each make and model, then displayed the top 10 for each category. The plot of vehicle makes categorizes the type of electric vehicle, while the model plot is not. The stacked EV type is done to contribute to the overall goal of finding the difference in use between the two. Descriptive counts were used to address the sub question about what makes and models were most common. In question 2 we created a bar chart of vehicle counts by model year, again split by electric vehicle type. These stacked bars display quantity, and divide usage by the type of vehicle which is owned. In question 3, a simple linear regression of Electric Range on Model Year was used for PHEVs only. The hypothesis tested is whether the slope differs from zero to determine if the change in year is a significant predictor of the number of EVs. Although planned for both vehicle types, BEV electric range values are zeroed out in recent model years in the WA DOL data and were therefore excluded. Year was centered at 2010 to make the intercept able to be interpreted as the predicted PHEV range in that year. A residual plot was generated to assess the linear model’s assumptions of constant variance and zero-mean errors.

Question 4 extends the initial three questions by digging deeper into external data and different methods. For the first part of this question, EV counts per ZIP code from the WA DOL dataset were joined to demographic estimates from the U.S. Census Bureau’s 2022 American Community Survey 5-year estimates, accessed through the tidycensus package. The response variable is EVs per 1,000 residents. Predictors are median household income, total population, and percent of residents holding a bachelor’s degree. ZIPs with fewer than 10 registered EVs were dropped, since low-count ZIPs produce unstable per-1,000 rate estimates that reflect noise rather than adoption patterns. A multiple linear regression was fit, and a residual plot was created from it. For the second part of question 4, a k-nearest neighbors classifier was implemented to predict electric vehicle type from Model Year and Legislative District, using a 70/30 train/test split. The default caret k grid ($k = 5, 7, 9$) was searched. KNN was used to test whether geography and model year alone carry enough significance to distinguish BEVs from PHEVs.

The software used was R 4.5 with RStudio, and the analysis relied on the R packages tidyverse for data manipulation and visualization, broom for tidying model output, caret for KNN training and confusion matrix

evaluation, and tidycensus for retrieving Census ACS data.

3 Results

3.1 Most common manufacturers and models

Tesla dominates the Washington EV fleet, holding 50.0% of registrations across the top 10 manufacturers. The next nine manufacturers combined, Chevrolet (8.4%), Nissan (7.0%), Ford (6.7%), Kia (6.1%), Toyota (5.2%), BMW (5.1%), Hyundai (4.5%), Rivian (3.9%), and Volkswagen (3.2%), count for the remaining half. The stacked fill on the manufacturer plot shows that Tesla and Rivian are entirely BEV, while Toyota, Ford, BMW, and Kia carry visible PHEV share, indicating that powertrain strategy splits cleanly between newer pure-EV manufacturers and traditional automakers transitioning through hybrid offerings. The model plot reinforces the Tesla concentration: the Model Y (38.1%) and Model 3 (24.2%) together account for over 60% of the top 10 models, with the Nissan Leaf (8.6%) the only non-Tesla model in third place.

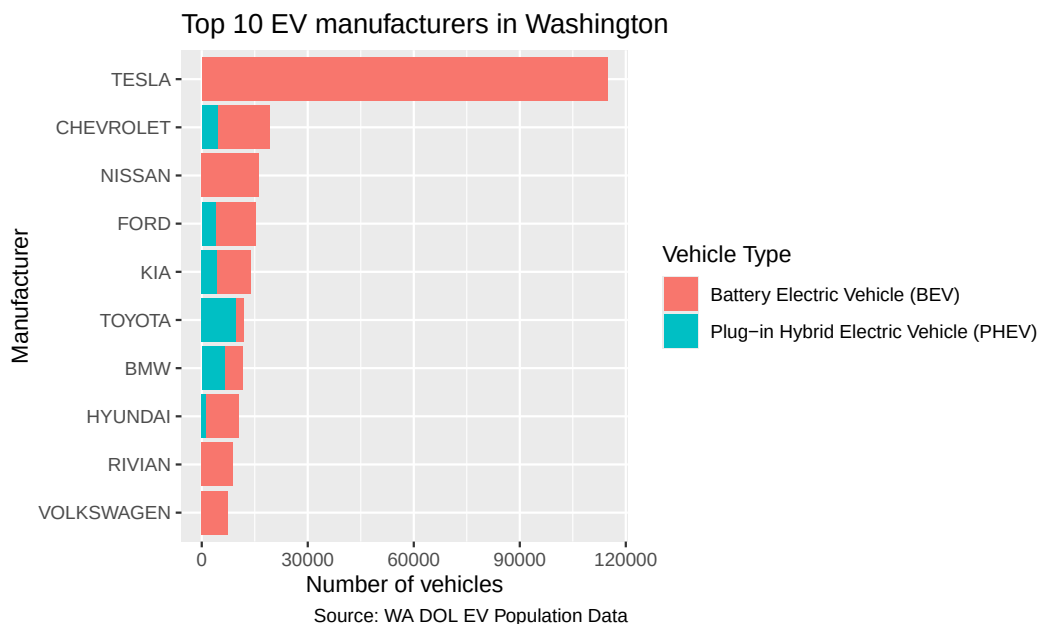


Figure 1: Top 10 EV manufacturers in Washington, filled by EV type.

3.2 BEV vs PHEV composition over model years

Total EV volume in Washington grew sharply through 2023, with the largest model-year cohort exceeding 60,000 vehicles. The BEV share grew each year while the PHEV share decreased in proportion to the whole, consistent with consumer transition in the market. We expect that the drop in 2025 and 2026 reflects partial-year registrations, since vehicles from these model years are still being registered as the dataset snapshot date precedes their full sales cycle. The trend confirms the expectation set in the introduction, that BEVs have become the dominant powertrain choice for new EV buyers in Washington.

3.3 PHEV electric range trend over model years

PHEV electric range has increased by approximately 0.34 miles per model year ($p = 2.04 \times 10^{-8}$), a statistically significant but practically negligible trend. The R^2 of 0.0076 indicates that Model Year alone explains less than 1% of the variation in PHEV electric range. Within-year variation across vehicle models dominates over any year to year movement. The residual plot reinforces this, as residuals show clear heteroscedasticity, with spread increasing at higher fitted values, and a cluster of large positive residuals at

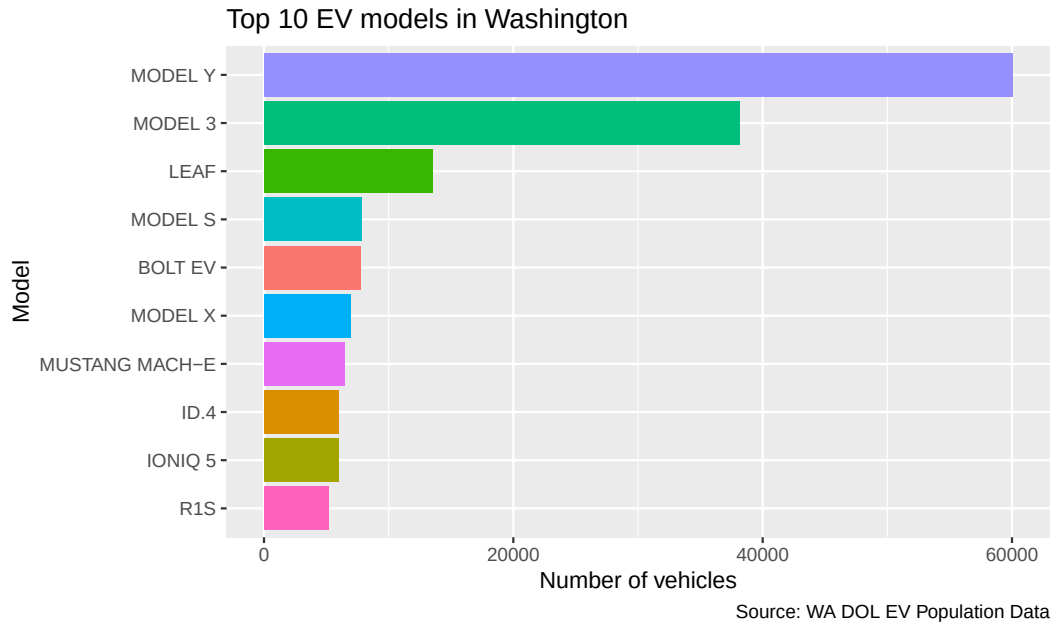


Figure 2: Top 10 EV models in Washington.

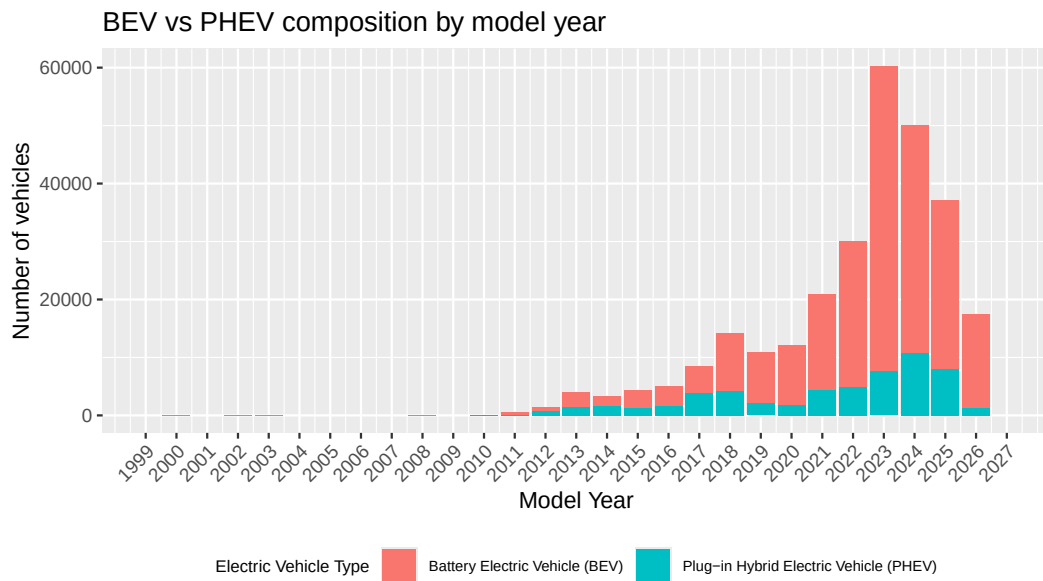


Figure 3: BEV vs PHEV composition by model year. Note that the most recent model years are partial-year snapshots.

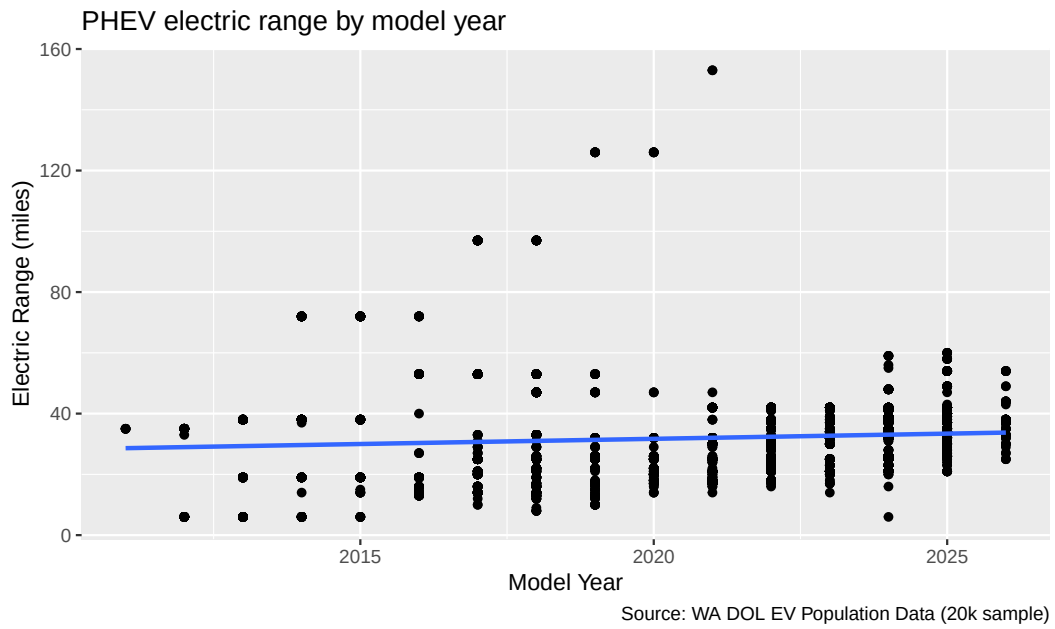


Figure 4: PHEV electric range by model year with linear fit.

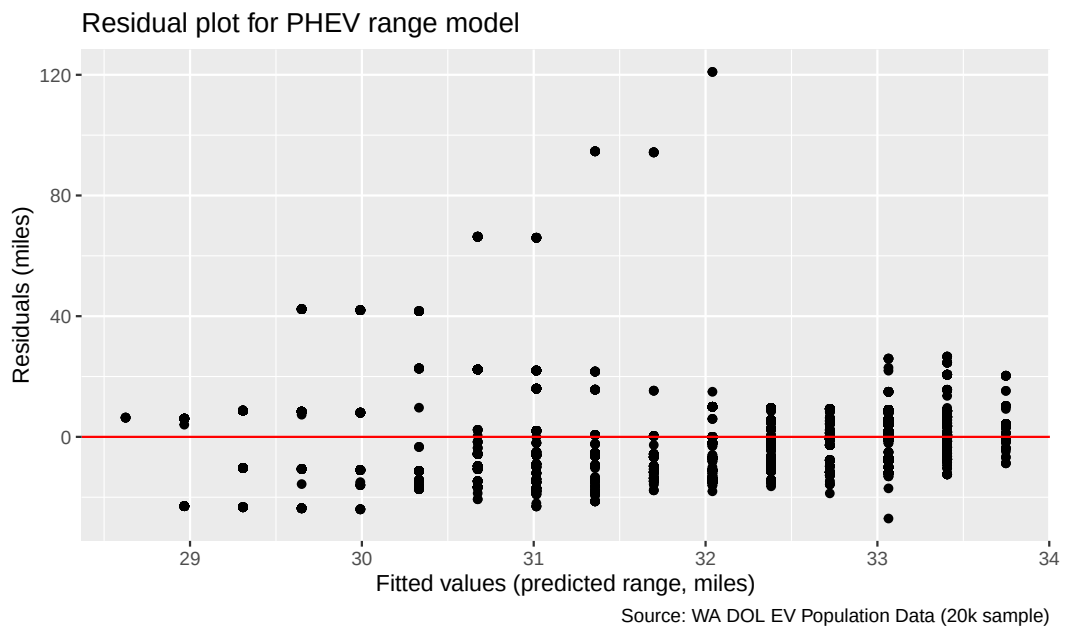


Figure 5: Residual plot for the PHEV range linear model.

fitted values 30–32 miles corresponding to the BMW i3 REx outliers. The slope is therefore reliable as a direction indicator but not as a predictive tool, and the model suggests that PHEV product diversification has outpaced per-vehicle range improvement.

3.4 Demographic predictors of EV adoption

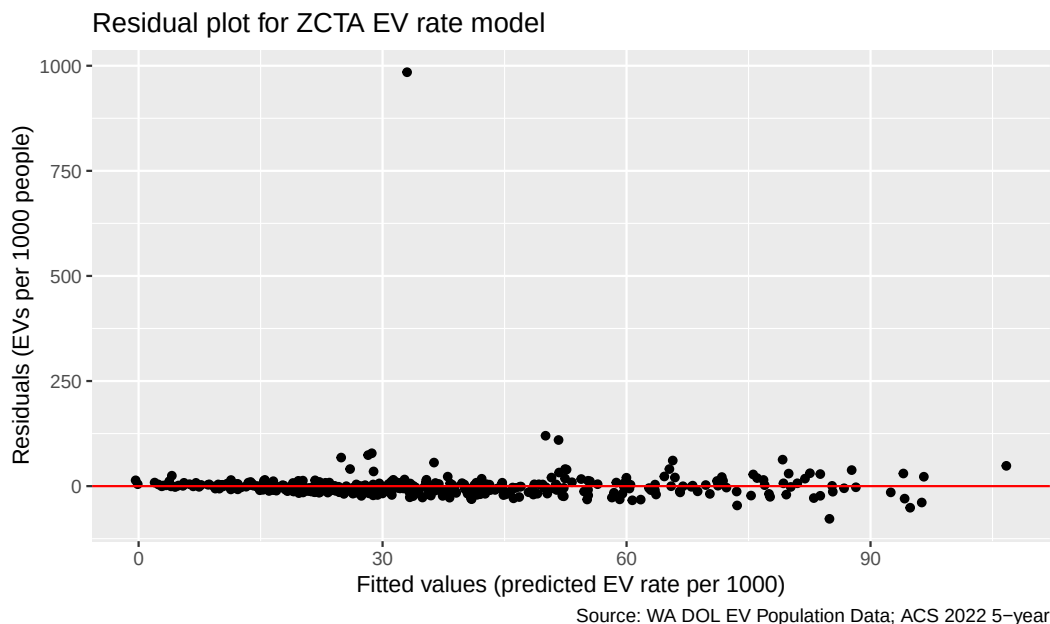


Figure 6: Residual plot for the multiple linear regression of EV rate per 1,000 residents on ZCTA-level demographics.

Education is the strongest demographic predictor of EV adoption in Washington. Each one-percentage-point increase in the share of residents holding a bachelor’s degree is associated with 2.20 more EVs per 1,000 residents ($p = 2.13 \times 10^{-7}$), holding income and population size constant. Median household income contributes independently but with smaller effect: each additional \$10,000 of income corresponds to roughly 2.7 more EVs per 1,000 residents ($p = 0.009$). Total population is not a significant predictor ($p = 0.13$), confirming that population size alone does not drive per-capita adoption. The model’s R^2 of 0.149 indicates that these three demographic variables together explain about 15% of the variation in adoption rates across ZIP codes, leaving the majority of variation driven by factors not measured here, such as charging infrastructure, housing type, or employer incentive programs. The residual plot shows the model fits well across most ZIP codes, with a small number of high-residual outliers retained from the `ev_count >= 10` filter.

3.5 Classifying EV type from geography and year

The KNN classifier reached a test accuracy of 79.6% ($k = 9$), only marginally above the no-information baseline of 79.4% that would be achieved by always predicting BEV. The kappa statistic of 0.12 confirms that agreement beyond chance is minimal. The confusion matrix shows the model correctly identified 4,634 of 4,759 BEVs (sensitivity 0.97) but only 140 of 1,236 PHEVs (specificity 0.11). This is a null finding, that Model Year and Legislative District alone do not carry enough signal to distinguish BEVs from PHEVs. The interpretive value of this result is that BEV and PHEV adoption is geographically mixed across Washington, rather than concentrated in identifiable regions or eras.

4 Conclusion

The strongest findings concern market composition. Tesla dominates the Washington EV market, holding half of all registrations among the top 10 manufacturers; the BEV share has grown each year while PHEVs have contracted; and education is the strongest demographic predictor of adoption rate, ahead of income and unrelated to raw population size. Two findings were null or weak. PHEV electric range has technically improved by 0.34 miles per model year, but with $R^2 < 0.01$ the trend explains almost none of the variation between vehicles, indicating that within-year diversity across PHEV models matters far more than any year-over-year improvement. The KNN classifier failed to beat the majority-class baseline, confirming that BEV/PHEV adoption is geographically mixed rather than regionally specialized.

Several methodological weaknesses limit the strength of these conclusions. The Q3 residuals show clear heteroscedasticity and a non-Gaussian shape, so the linear model is appropriate only as a direction indicator and not as a predictive tool; a robust regression or year-binned summary would be more defensible. The Q4a model achieves an R^2 of 0.149, which is useful for direction-finding but not for prediction; the demographic predictors collectively miss roughly 85% of the variation in adoption rates. The Q4b classifier was evaluated by accuracy when a balanced metric such as kappa or F1 would have shown its near-baseline performance more directly; reporting accuracy without baseline comparison would have overstated the result.

The reliability and validity of the data are mixed. The dataset’s main strength is that it covers the full population of registered EVs in Washington rather than a sample, removing sampling concerns. Its weaknesses are that it is a registration snapshot — no purchase timing, no flow, no scrappage, and no out-of-state buyers — and that BEV electric range is no longer maintained by WA DOL, which forced the Q3 PHEV-only restriction. The ACS data used in Q4a covers 2018–2022, slightly lagging the EV registration data, which introduces a timing mismatch.

The findings are also limited by scope. They apply to Washington only and should not be assumed to generalize to other states with different infrastructure, climate, or incentive policies. Future work could pair this data with charging-station density to test infrastructure as an adoption driver, perform time-series analysis once multiple registration snapshots are available, or compare Washington to Oregon and California using equivalent state-DOL datasets.

5 References

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